Effect of Caffeine on Locomotor Activity and Sleep in Drosophila melanogaster

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Introduction

Drosophila melanogaster (fruit flies) enters a sleep state that depends on time of day and rebounds after sleep deprivation (Hendricks et al., 2000; Shaw et al., 2000). In this experiment, sleep is defined experimentally as five or more consecutive minutes without locomotor activity, allowing both activity and sleep to be measured automatically using the Drosophila Activity Monitor (DAM) system. The DAM system records one count each time a fly crosses an infrared beam at the midpoint of its tube. A long period of inactivity is labeled as a sleep bout.

Caffeine is a well-known stimulant that is used in everyday life. It is known to increase wakefulness; however, the way it does so in Drosophila differs from that in humans. In fruit flies, caffeine keeps them awake through protein kinase A signaling rather than via the adenosine receptor (Wu et al., 2009).

This experiment measured how caffeine affected activity and sleep in fruit flies over 7 days. The hypothesis was that flies fed caffeine would show greater total activity and shorter sleep periods, and that these effects would differ between males and females due to differences in body weight.

Materials and Methods

Materials

Drosophila melanogaster (fruit flies) were each individually put in clear glass tubes sealed with a food plug and wax at one end and a plastic cap at the other, and colored marking bands were used to identify treatment groups. The scientist separated flies by sex under a fluorescent microscope, using CO_2 as an anesthetic. A fine paintbrush was used to move anesthetized flies into the tubes.

For the food, Caffeine powder was massed on an analytical balance using weigh paper and mixed into standard fly food.

The Fruit Flies activity was recorded with a TriKinetics Drosophila Activity Monitor, which has 32 tubes and passes an infrared beam across the midpoint of each tube. The monitor was connected to a computer running DAMSystem3 data collection software, which was set to record total beam-break counts in one-minute bins.

Food Preparation

The solution used in the food was prepared by dissolving 1 g of caffeine in 200 mL of water, yielding a concentration of 5 mg/mL. For the caffeine food, approximately 13.932 g of fly food was mixed with 69.7 mL of liquid per batch, consisting of 7 mL of caffeine solution diluted in

62.7 mL of water. Each patch, therefore, had around 35 mg of caffeine, yielding a final caffeine concentration of approximately 0.5 mg/mL, which was fed to the flies by mixing it into their food.

Loading of Flies

Flies were anesthetized and transferred individually into the glass tubes with a paintbrush, one fly per tube. The scientist then sealed each tube so that the fly could move the full length of the tube and had continuous access to food for the duration of the recording. All 32 positions in the monitor were filled, yielding 8 flies per group: male control, male caffeine, female control, and female caffeine.



Figure 1: Preparation of tubes and food: Glass tubes are laid out on the bench after being filled. The white rack at the back center is the TriKinetics monitor, and the small black plastic caps used to seal the open end of each tube are scattered across the bench.

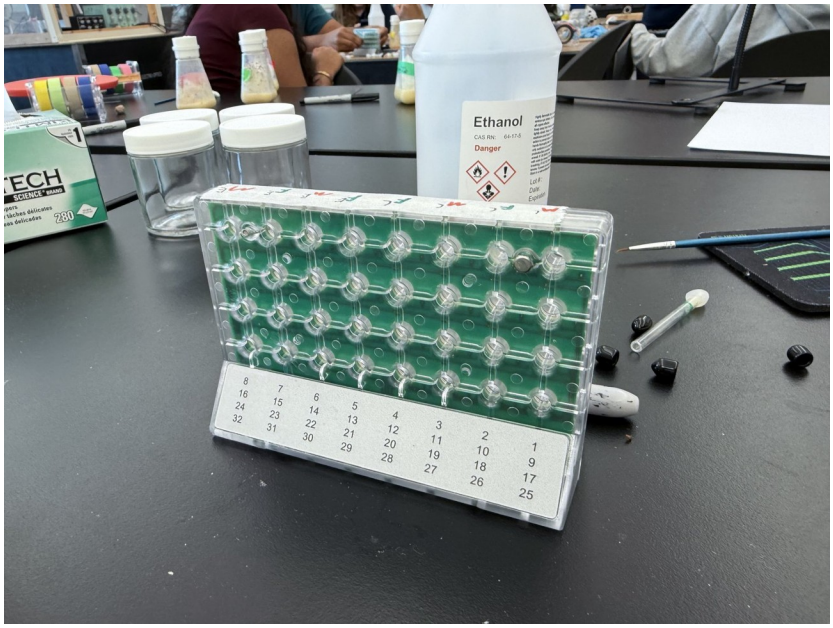


Figure 2: Loaded DAM monitor: This is the monitor used in the experiment. It holds 32 tubes in eight columns of four. The scientist placed a paper strip, taped across the top of the monitor, recording the sex of each fly in each column, with M for male and F for female, showing the alternating arrangement described above.

Data Collection Procedures

The monitor was placed on a table next to a window, so the flies were exposed to natural light conditions at Jupiter, Florida.

The light cycle was recorded using the DAM's built-in light sensor and verified against publicly available weather data on daylight to confirm that the sensor was tracking real daylight (Table 1).

Data Analysis

The first 50 minutes of the recording were discarded because flies were still being loaded, so analysis began at 16:00 on 17 September.

Sleep was counted as five or more consecutive minutes in which the tube recorded zero counts (Pfeiffenberger et al., 2010b). Consecutive sleep minutes were then grouped into sleep bouts, and the number of bouts and their lengths were counted for each fly. Each minute was classified as day or night based on the light sensor reading for that minute, rather than a fixed clock time.

A fly was considered dead once it stopped moving for the remainder of the experiment. Its time of death was recorded as the last hour in which it showed clear activity, defined as at least 3 movement counts, to avoid treating accidental movement occurring in the lab as fly movement. This data was also checked against the researcher's observations of which flies were alive on day 7 of the experiment.

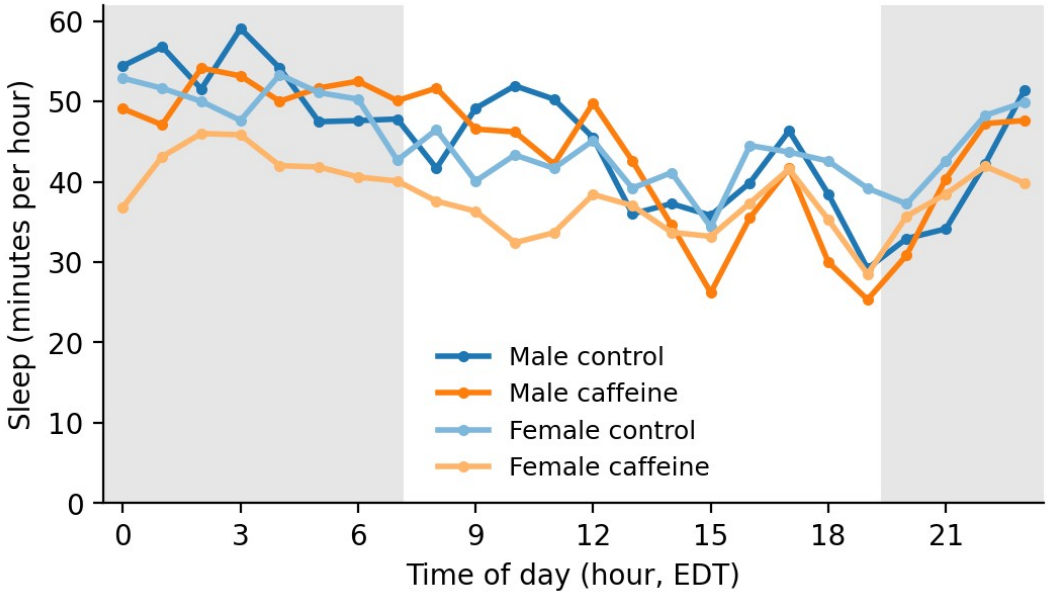
Group values were reported as the mean of the eight flies in that group plus or minus the standard deviation.

With only eight flies per group and most dying before the recording ended, the sample was too small for any meaningful statistical test to be carried out; therefore, only descriptive statistics were calculated. Differences between caffeine and control groups are therefore described in terms of the means and their spread, and should be read as trends rather than as proven effects.

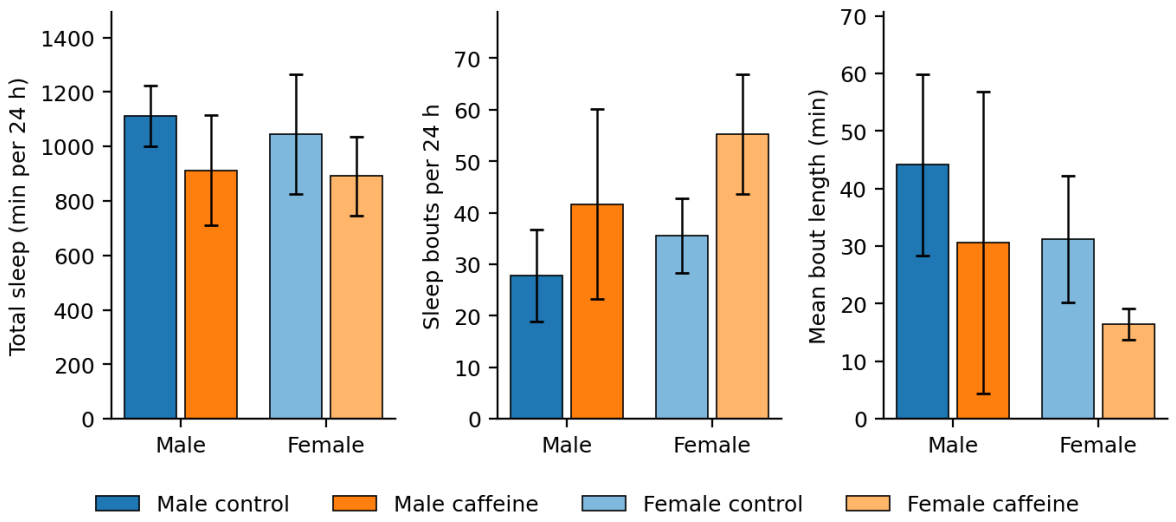
Data/Results

Table 1: Mean and standard deviation for each group: Values are the mean of the eight flies in each group plus or minus the standard deviation.

Measure	Male control	Male caffeine	Female control	Female caffeine
Total sleep (min/24 h)	1112 ± 113	913 ± 203	1045 ± 221	891 ± 145
Sleep bouts (per 24 h)	27.8 ± 8.9	41.7 ± 18.4	35.6 ± 7.3	55.3 ± 11.7
Mean bout length (min)	44.1 ± 15.8	30.6 ± 26.3	31.2 ± 11.0	16.4 ± 2.7
Longest bout (min)	431 ± 181	311 ± 361	491 ± 314	121 ± 58
Activity (counts/24 h)	669 ± 305	833 ± 471	666 ± 351	1074 ± 332
Night activity (counts/h)	22.3 ± 8.6	33.0 ± 21.1	22.3 ± 16.9	43.6 ± 20.5
Median survival (h)	122.7	40.0	74.5	39.2



Graph 1: Sleep across the day in each treatment group: The x-axis shows the time of day in hours, and the y-axis shows the mean sleep minutes per hour. Each point is the average across all minutes in that clock hour over the first 48 hours of recording. This window was used because most flies were alive in each group during that time; specifically, the means were 7.5, 6.1, 7.8, and 6.2 flies per point for male control, male caffeine, female control, and female caffeine, respectively. The light gray shows periods of no light, and the white period indicates periods of light.



Graph 2: Sleep measures in each treatment group: The bars show the group means, and error bars show the standard deviation. The left panel shows total sleep in minutes per 24 hours, the center panel the number of sleep bouts per 24 hours, and the right panel the mean length of a sleep bout in minutes.

Written Results

Caffeine produced more frequent but shorter sleep bouts (Table 1; Graph 2). Bout number increased from 27.8 ± 8.9 to 41.7 ± 18.4 in males and from 35.6 ± 7.3 to 55.3 ± 11.7 in females, and the mean bout length decreased from 44.1 ± 15.8 to 30.6 ± 26.3 minutes in males and from 31.2 ± 11.0 to 16.4 ± 2.7 minutes in females. The total amount of sleep changed a lot less from 1,112 ± 113 to 913 ± 203 minutes per 24 hours in males, and from 1,045 ± 221 to 891 ± 145 minutes in females.

Caffeine increased activity in females, from 666 ± 351 to 1,074 ± 332 counts per day, while males showed little to no change (Table 1). All four groups kept a daily rhythm, sleeping more during the dark hours than the light hours, but the caffeine groups slept less across the day than their controls (Graph 1).

Activity for each fly was substantially different within groups (Appendix A). Caffeine also reduced survival, with median survival falling from 122.7 to 40.0 hours in males and from 74.5 to 39.2 hours in females.

Discussion

This experiment tested whether caffeine alters activity and sleep in Drosophila melanogaster and whether the sexes respond differently. Caffeine altered sleep in both sexes, increasing the number of sleep bouts and decreasing their duration, with total sleep decreasing only slightly.

The sexes responded differently to caffeine. Females became more active, while males showed a smaller increase in activity despite having more periods of shorter sleep. This difference may be attributable to body mass, since females are larger, but both sexes were given the same concentration of food.

The biggest limitation of this experiment was the high mortality rate. Caffeine reduced the survival, but mortality was still unusually high in the controls. Furthermore, uncontrolled natural environments were another variable that could have affected the results. Future experiments should use larger sample sizes and control light and overall environmental conditions.

Overall, the hypothesis was partly supported: caffeine consistently led to more sleep periods, but they were shorter in both sexes. Furthermore, the predicted increase in total activity appeared only in females.

References

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Appendix A: Individual Fly Data

Here is the raw data for all 32 individual flies. Each row is one fly, identified by its tube number in the monitor. Survival is the time in hours from the start of analysis at 16:00 on 17 September until the fly last recorded activity.

Tube	Sex	Treatment	Survival (h)	Total sleep (min/24 h)	Sleep bouts (per 24 h)	Mean bout (min)	Longest bout (min)	Activity (counts/24 h)
1	Male	Control	151.2	1020	13.8	73.9	353	682
3	Male	Control	164.6	1265	25.5	49.7	436	329
9	Male	Control	164.7	1100	31.5	34.9	595	585
11	Male	Control	70.5	1105	22.1	49.9	385	582
17	Male	Control	25.4	1121	43.4	25.8	216	368
19	Male	Control	65.0	1255	24.4	51.5	773	611
25	Male	Control	94.3	922	34.6	26.6	257	1271
27	Male	Control	163.1	1109	27.2	40.8	430	920
6	Male	Caffeine	104.1	816	8.8	93.2	1127	873
8	Male	Caffeine	135.6	1268	44.9	28.3	374	259
14	Male	Caffeine	41.2	1025	40.2	25.5	455	681
16	Male	Caffeine	29.6	1060	59.1	17.9	68	423
22	Male	Caffeine	10.6	619	67.8	9.1	31	1833
24	Male	Caffeine	38.9	805	50.0	16.1	146	797
30	Male	Caffeine	29.7	770	33.1	23.2	113	999
32	Male	Caffeine	94.2	938	29.5	31.7	172	799
2	Female	Control	119.2	1150	31.9	36.0	401	396
4	Female	Control	47.9	875	47.6	18.4	188	828
10	Female	Control	83.5	1256	34.5	36.4	780	335
12	Female	Control	66.1	826	41.7	19.8	217	928
18	Female	Control	62.5	946	28.8	32.8	493	907
20	Female	Control	37.8	753	41.9	18.0	108	1231
26	Female	Control	83.4	1219	27.6	44.2	859	408
28	Female	Control	82.8	1335	30.4	43.9	880	291
5	Female	Caffeine	24.7	979	65.0	15.1	216	798
7	Female	Caffeine	28.7	949	68.5	13.9	45	943
13	Female	Caffeine	20.8	1023	69.2	14.8	69	859
15	Female	Caffeine	33.1	656	44.3	14.8	92	1823
21	Female	Caffeine	164.6	1008	45.7	22.0	178	977
23	Female	Caffeine	81.7	1001	60.2	16.6	100	860
29	Female	Caffeine	54.5	763	41.0	18.6	156	1147
31	Female	Caffeine	45.3	747	48.2	15.5	109	1183